STORE LINK: CONNECTS LOCAL STORE WITH ONLINE CUSTOMERS

A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled **"Store Link: Connects Local Store With Online Customers"** is a record of Bonafide work of team members **Chida Nikith – 2320090023, Koneru Nikhil – 2320090025, and Koduru Vihaari – 2320090027**, submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled **"Store Link: Connects Local Store With Online Customers"** is a Bonafide work carried out and submitted by **Chida Nikith – 2320090023, Koneru Nikhil – 2320090025, and Koduru Vihaari – 2320090027**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

Signature of the Supervisor

Signature of the HOD

Signature of the External Examiner

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TABLE OF CONTENTS

S.No	Contents	Page no
1	Abstract	6
2	Introduction	7
3	Literature survey	9
4	Hardware and Software requirements	11
5	Implementation	13
6	Experimentation and Code	16
7	Results	18
8	Conclusion	22
9	References	23

ABSTRACT

The *Store Link: Connecting Local Stores with Online Customers* project is an innovative digital solution aimed at bridging the gap between traditional retail stores and modern online shopping platforms. With the rapid growth of e-commerce, many small-scale local businesses struggle to compete due to the lack of digital presence and technological infrastructure. This project addresses that challenge by providing a full-stack web application that enables local kirana stores to sell their products online while allowing customers to conveniently browse, select, and order items from nearby stores.

The system is developed using modern web technologies including React for the frontend, Node.js and Express for backend services, and MongoDB for database management. It provides essential features such as user authentication, product management, cart functionality, and order placement. Additionally, advanced features like WhatsApp-based order notification, responsive user interface design, and Swiggy-style address management enhance user experience and operational efficiency.

The application workflow ensures seamless interaction between users and store owners, where customers can place orders that are stored in the database and simultaneously communicated to the store via WhatsApp. This reduces manual effort and improves communication speed. The system is designed to be scalable, user-friendly, and adaptable for real-world applications.

Overall, the project demonstrates how digital innovation can empower local businesses, improve customer convenience, and promote the adoption of technology in the retail sector. It provides a cost-effective and efficient solution that supports the growth of small businesses in the competitive e-commerce environment.

INTRODUCTION

In recent years, the retail industry has undergone a significant transformation due to the rise of digital technologies and e-commerce platforms. Large companies such as Amazon and Flipkart have dominated the online shopping space, offering customers convenience, variety, and quick delivery services. However, small local stores, commonly known as kirana stores, often lack the resources and technical knowledge to establish an online presence, making it difficult for them to compete in this digital era.

The *Store Link* project is developed as an innovative solution to address this issue by connecting local stores with online customers through a simple and efficient web application. The system enables customers to browse available products, add items to a cart, and place orders without physically visiting the store. At the same time, it allows store owners to manage their products and orders digitally, improving operational efficiency.

The project leverages modern full-stack development technologies to create a responsive and scalable platform. It also integrates communication features such as WhatsApp notifications, ensuring that store owners receive real-time updates about customer orders. This approach not only improves convenience for customers but also helps local businesses expand their reach and increase sales.

By combining traditional retail with modern digital solutions, the *Store Link* system contributes to the digital transformation of small businesses and supports the development of a more inclusive and accessible e-commerce ecosystem.

Problem Description:

Despite the rapid growth of online shopping, many local retail stores still operate using traditional methods, which limits their ability to reach a wider customer base. The absence of digital platforms for these stores creates several challenges for both store owners and customers.

Local store owners face difficulties in managing inventory, tracking orders, and communicating with customers efficiently. Since they rely on physical interactions, their business operations are often time-consuming and less organized. Additionally, they are unable to compete with large e-commerce platforms that offer advanced features such as online ordering, quick delivery, and digital payments.

On the other hand, customers experience inconvenience as they must visit stores physically to purchase items. This becomes particularly challenging during busy schedules or situations like

pandemics, where contactless shopping is preferred. There is also a lack of real-time information about product availability, pricing, and delivery options.

Furthermore, there is no standardized system for integrating local stores into the digital ecosystem, which creates a gap between traditional retail and modern e-commerce solutions. This problem highlights the need for an innovative platform that can digitize local stores, streamline operations, and provide customers with a seamless shopping experience.

Objectives:

- To connect local stores with online customers
- To provide an easy product browsing and ordering system
- To enable cart and order management
- To implement user authentication
- To help store owners manage products and orders digitally
- To provide real-time order notifications
- To improve convenience and accessibility for users
- To support digital transformation of local businesses

LITERATURE SURVEY

The rapid growth of e-commerce platforms has transformed the way people purchase goods and services. Large platforms like Amazon and Flipkart provide customers with a wide range of products, fast delivery, and convenient online shopping experiences. However, these platforms are mainly focused on large-scale sellers, and small local stores often struggle to compete due to lack of digital infrastructure.

Research shows that digital transformation in retail significantly improves business performance and customer satisfaction. By adopting online systems, businesses can expand their reach, manage inventory efficiently, and provide better services. However, many local kirana stores still rely on traditional methods, which limits their growth and accessibility.

Several applications like Swiggy and Zomato have introduced quick delivery and location-based services, which inspired the address management and delivery concepts used in this project. These platforms demonstrate how technology can simplify ordering and improve user experience.

Modern web technologies such as React, Node.js, and MongoDB have enabled the development of scalable and efficient web applications. These technologies support real-time data handling, fast performance, and easy integration of features like authentication and notifications.

Despite these advancements, there is still a gap in solutions specifically designed for small local stores. This project addresses that gap by providing a simple, cost-effective, and scalable platform that connects local stores with online customers, thereby improving accessibility and supporting digital business growth.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

The *Store Link* system is a web-based application, and most of the processing is handled through the backend server and cloud database. Therefore, high-end hardware is not required, but a basic system configuration is necessary for smooth development and usage.

Processor:

A system with an Intel Core i3 processor or equivalent is sufficient for running the application. For better performance during development and testing, an Intel Core i5 or higher is recommended.

RAM:

A minimum of 4 GB RAM is required to run the development environment, browser, and application efficiently. However, 8 GB RAM is recommended for smoother multitasking and faster performance.

Storage:

At least 500 MB to 1 GB of free disk space is required for storing project files, dependencies, and temporary data.

Graphics:

No dedicated graphics card is required. Integrated graphics are sufficient since the project focuses on web development and not graphical processing.

Operating System:

The system can run on Windows, Linux, or macOS, as the application is platform-independent and accessed through a web browser.

Software Requirements

The project uses modern web development technologies and tools for building a scalable and efficient system.

Frontend Technologies:

- React – Used for building a responsive and interactive user interface

Backend Technologies:

- Node.js – Handles server-side logic

- Express.js – Used for creating APIs and routing

Database:

- MongoDB – Stores user data, products, and orders

Development Tools:

- Visual Studio Code – Used for coding and development
- Web Browser (Google Chrome / Microsoft Edge) – Used to run and test the application

Additional Tools:

- WhatsApp – Used for sending order notifications
- Local Storage – Used for storing user address data

Conclusion of Requirements

With the above hardware and software configuration, the system can be developed and executed efficiently. Since it is a web-based application, it ensures flexibility, scalability, and minimal dependency on high-end hardware while providing a smooth user experience.

IMPLEMENTATION

The *Store Link: Connecting Local Stores with Online Customers* system is implemented as a full-stack web application that provides individual websites for local kirana stores. The system enables store owners to digitize their business operations while allowing customers to access products and place orders online. The implementation follows a modular and scalable architecture consisting of frontend, backend, and database components.

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1. System Architecture

The system is designed using a three-tier architecture:

- **Frontend Layer:** Developed using React, which provides an interactive and responsive user interface. It allows users to browse products, manage carts, and place orders easily.
- **Backend Layer:** Built using Node.js and Express.js, which handle server-side logic, API requests, authentication, and communication between frontend and database.
- **Database Layer:** Managed using MongoDB, which stores user information, product details, order data, and store-specific data.

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2. Store-Based Website System

A key feature of this project is that each kirana store is provided with its own dedicated website interface. This allows:

- Independent product management for each store
- Customized product listings
- Separate order handling for each store

This approach ensures that every store operates digitally while maintaining its individuality.

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3. User Authentication System

The system includes a secure authentication mechanism:

- Users can register and log in to the platform

- Credentials are validated through backend APIs
- User data is securely stored in the database

This ensures safe access to user accounts and personalized shopping experience.

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4. Product Management System

Store owners are provided with features to manage their products:

- Add new products with details such as name, price, and category
- Update existing product information
- Delete unavailable products

This helps in efficient inventory management and keeps product listings updated.

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5. Cart and Order Management

The system provides a complete shopping workflow:

- Customers can browse products and add them to the cart
- Modify quantity or remove items from the cart
- Place orders after confirming details

Once the order is placed, it is stored in the database and made available to the store owner.

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6. Address Management System

The application includes a Swiggy-style address input system:

- Users can enter and save their delivery address
- Address is stored using local storage
- Used during order placement for delivery

This improves convenience and reduces repeated data entry.

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7. WhatsApp Notification Integration

To enhance communication:

- Order details are sent via WhatsApp

- Store owners receive real-time notifications
- Helps in faster order processing

This feature bridges the gap between digital ordering and traditional store communication.

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8. System Workflow

The working of the system follows these steps:

1. User registers or logs into the system
2. Selects a local store website
3. Browses products and adds items to cart
4. Enters delivery address
5. Places the order
6. Order is stored in the database
7. Notification is sent to store owner via WhatsApp
8. Store owner processes and fulfills the order

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9. User Interface Design

The system is designed with a focus on usability:

- Responsive design for mobile and desktop devices
- Simple navigation for all user types
- Clean and intuitive interface

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10. Overall Implementation

The implementation successfully integrates modern web technologies to create a scalable and efficient platform. It provides a practical solution for digitizing local stores and connecting them with customers, ensuring improved accessibility, better communication, and enhanced business operations.

Experimentation and Code

The *Store Link* system was developed and tested as a full-stack web application using modern technologies. The experimentation process involved verifying the functionality of each module, ensuring smooth interaction between frontend, backend, and database, and testing the overall system workflow.

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1. Code Structure

The project follows a modular code structure divided into three main parts:

Frontend (React)

- Developed using React
- Responsible for user interface and interaction
- Includes components such as:
    - Login/Register pages
    - Product listing page
    - Cart page
    - Order page

Backend (Node.js & Express)

- Built using Node.js and Express.js
- Handles:
    - API requests
    - User authentication
    - Product management
    - Order processing

Database (MongoDB)

- Managed using MongoDB

- Stores:
    - User details
    - Product data
    - Order information

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2. Experimentation Steps

The following steps were performed during the development and testing phase:

1. Designed the frontend interface using React components
2. Developed backend APIs for handling requests and responses
3. Connected the frontend with backend using HTTP requests
4. Integrated MongoDB database for data storage
5. Implemented user authentication (Login/Register system)
6. Added product management features for store owners
7. Developed cart functionality for adding and removing products
8. Implemented order placement system
9. Integrated WhatsApp notification for order confirmation
10. Tested the complete workflow from product selection to order delivery

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3. Functional Testing

The system was tested under different scenarios to ensure reliability:

- Authentication Testing: Verified login and registration functionality
- Product Testing: Checked adding, updating, and deleting products
- Cart Testing: Ensured items can be added and removed correctly
- Order Testing: Verified successful order placement and storage
- Notification Testing: Confirmed WhatsApp messages are sent properly

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4. Performance Analysis

- Processing Speed: Fast response due to efficient API handling
- Scalability: Can support multiple stores and users simultaneously
- Data Handling: MongoDB efficiently manages large datasets
- User Experience: Smooth navigation and quick interactions

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5. Challenges Faced

During development, several challenges were encountered:

- Managing state across multiple pages in React
- Handling API errors and debugging backend issues
- Designing a responsive and user-friendly interface
- Ensuring proper integration between frontend and backend

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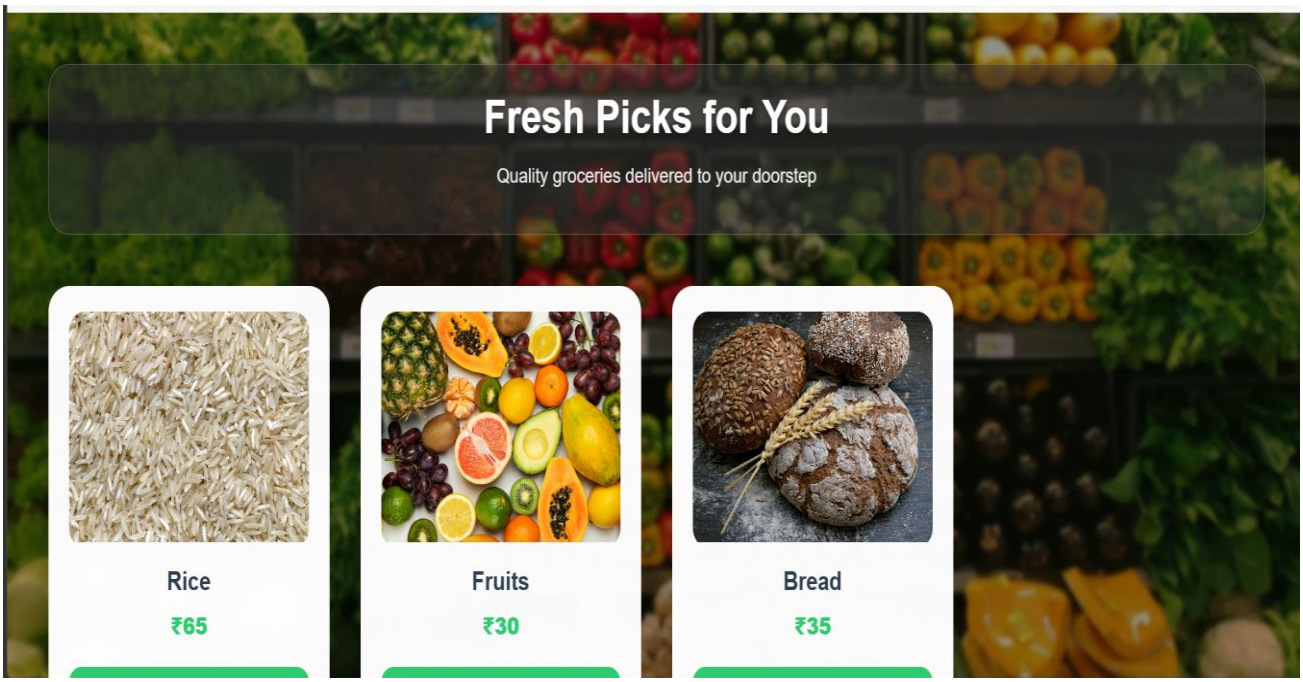
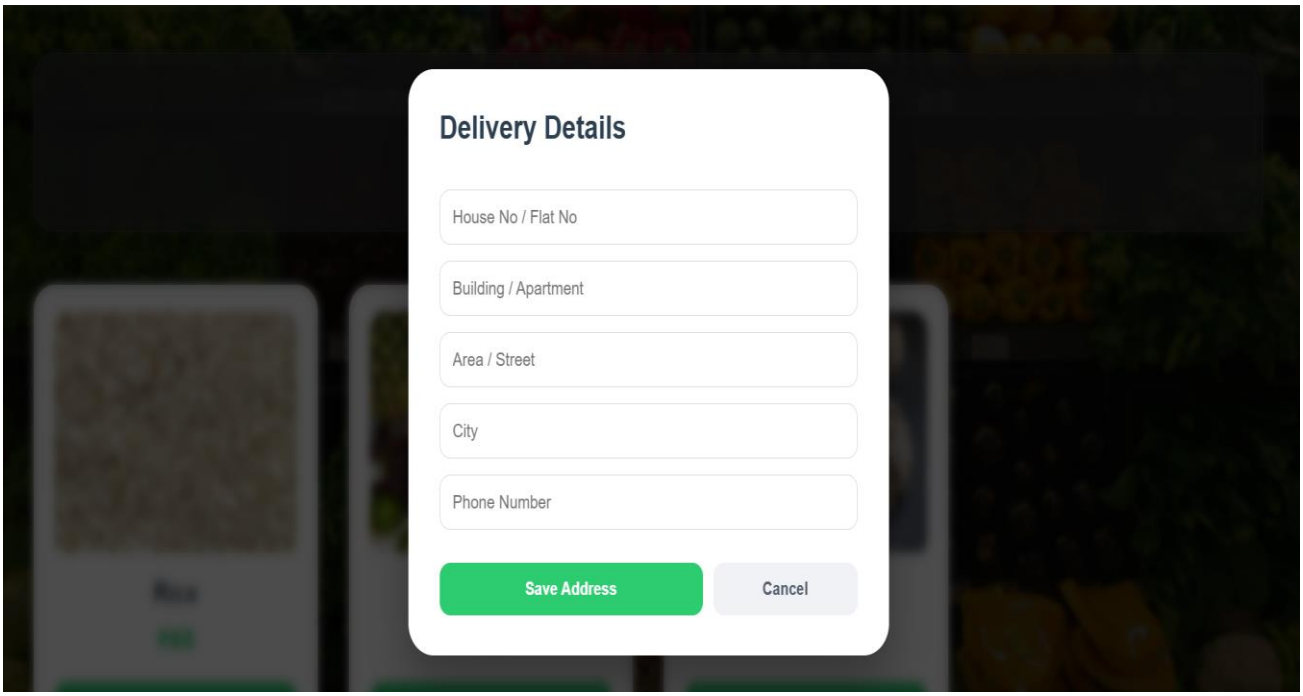
6. Conclusion of Experimentation

The experimentation phase confirmed that the system functions effectively and meets the project requirements. All modules, including authentication, product management, cart system, and order processing, were successfully implemented and tested. The integration of WhatsApp notifications enhanced communication between customers and store owners. Overall, the system proved to be reliable, efficient, and suitable for real-world applications.

RESULTS:


Fresh Picks for You
Quality groceries delivered to your doorstep
Rice
₹65
Fruits
₹30
Bread
₹35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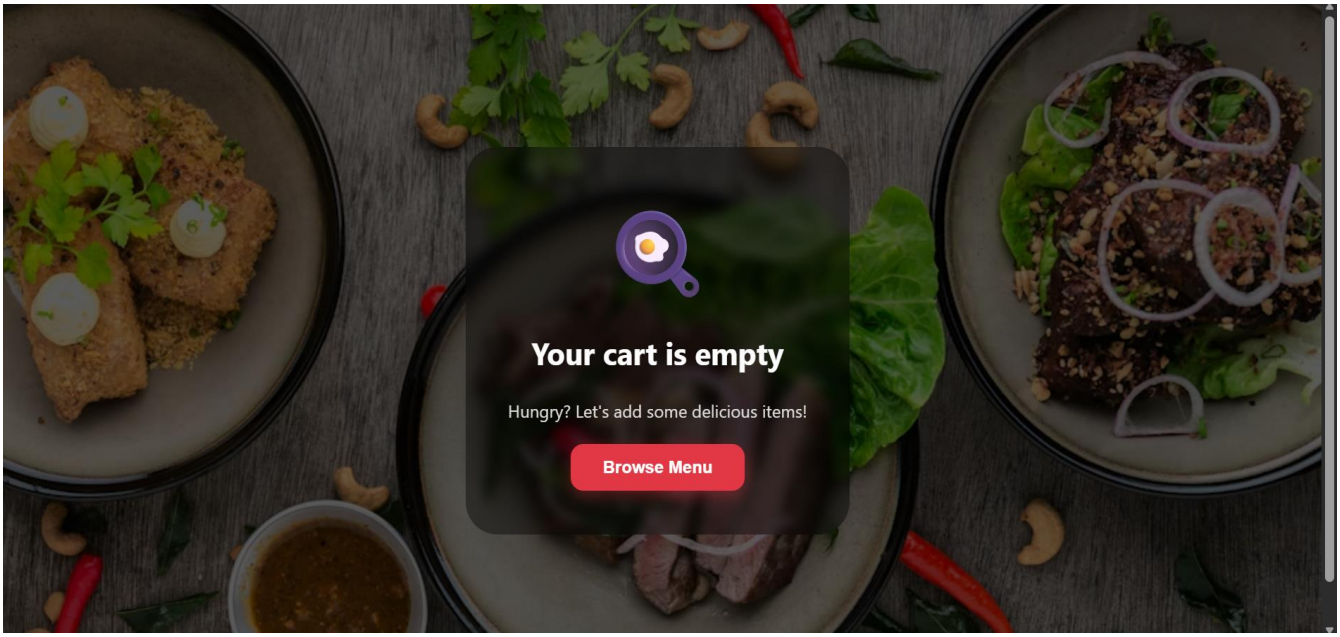
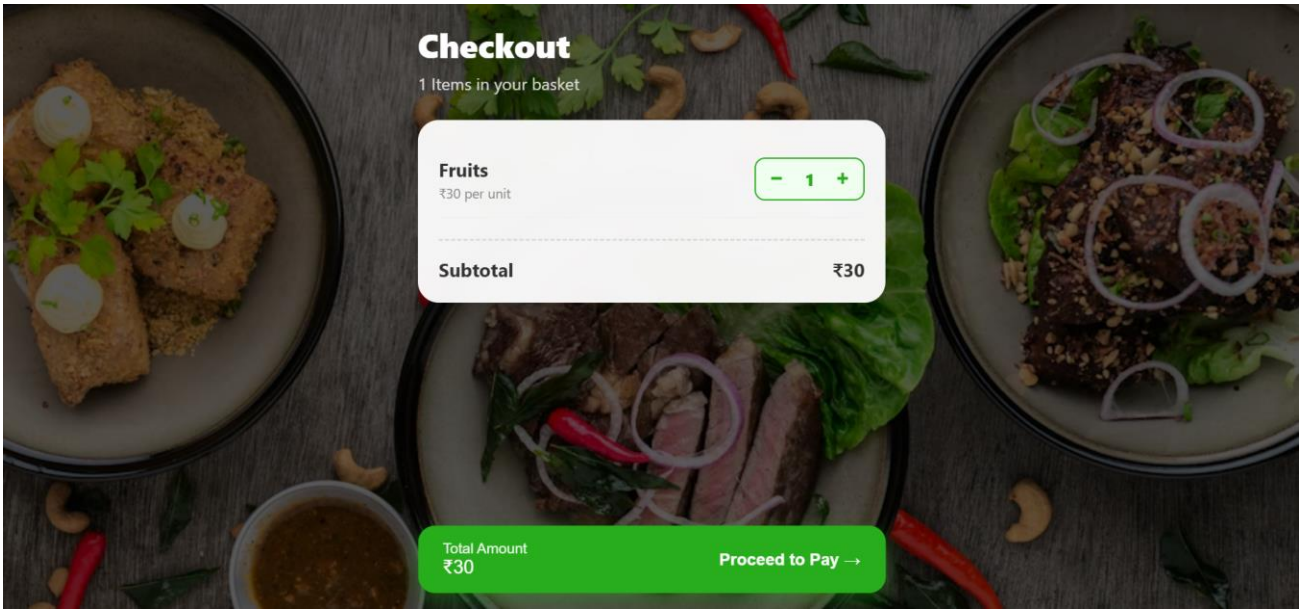
Building / Apartment
Area / Street
City
Phone Number
Save Address
Cancel


Your cart is empty
Hungry? Let's add some delicious items!
Browse Menu


Checkout
1 Items in your basket
Fruits
₹30 per unit
− 1 +
Subtotal
₹30
Total Amount
₹30
Proceed to Pay →

CONCLUSION

The *Store Link: Connecting Local Stores with Online Customers* project successfully demonstrates an innovative approach to digitizing local kirana stores by providing them with individual websites and connecting them with customers through an online platform. The system effectively bridges the gap between traditional retail businesses and modern e-commerce solutions.

The implementation of a full-stack web application using technologies such as React, Node.js, and MongoDB ensures a scalable, efficient, and user-friendly system. Features such as user authentication, product management, cart system, and order processing were successfully integrated, providing a smooth and seamless user experience.

One of the key achievements of this project is the integration of real-time communication using WhatsApp notifications, which enhances interaction between customers and store owners and reduces delays in order processing. The system also enables store owners to manage their inventory and orders digitally, improving operational efficiency.

The project highlights the importance of digital transformation for small businesses and demonstrates how technology can be used to improve accessibility, convenience, and business growth. By providing a simple and cost-effective solution, the system empowers local stores to compete in the digital marketplace.

In the future, the system can be enhanced by integrating online payment systems, order tracking features, multiple address management, and real-time notifications. These improvements can further increase the functionality and scalability of the platform.

Overall, the project achieves its objectives and provides a practical, scalable, and innovative solution for connecting local stores with online customers.

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